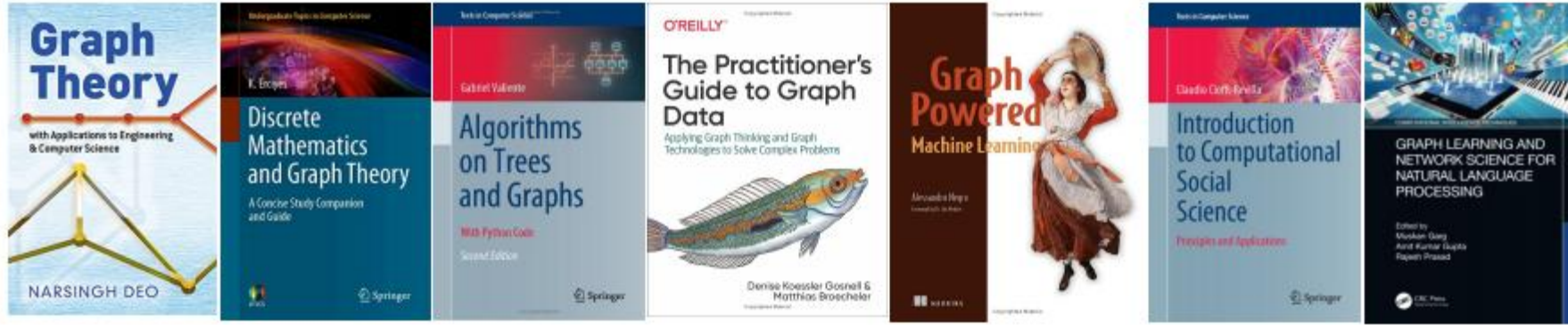


Chapter 10, Graphs

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- Spring 2026
- Aror University of Art, Architecture, Design & Heritage Sukkur



Graph Theory:

- Provide techniques for many clever data-structures for organizing information
- Many problems are modeled and efficiently solved using graphs
- particularly problems in Networks, Operating systems, Database

Graphs

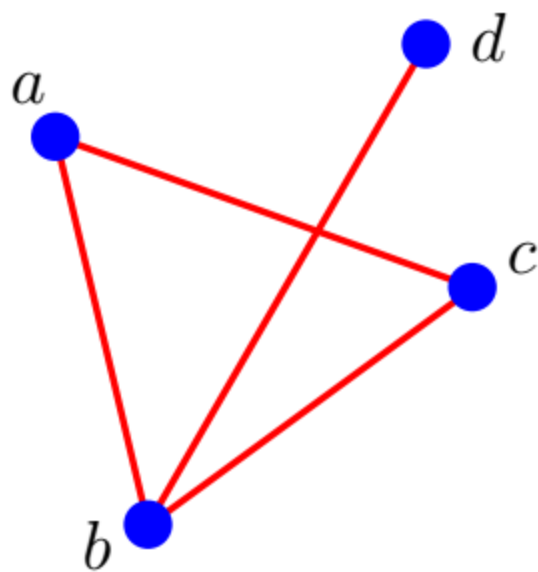
A graph is

- 1 a set of vertices V
- 2 a set of edges E ; each edge is a 2-subset of V

$$G = (V, E)$$

$$V = \{a, b, c, d\}$$

$$E = \{(a, b), (a, c), (b, c), (b, d)\}$$



Each edge is an unordered pair from V , we use the ordered pair notation

Types of Graphs: Undirected Simple Graphs

Applications require different types of graphs

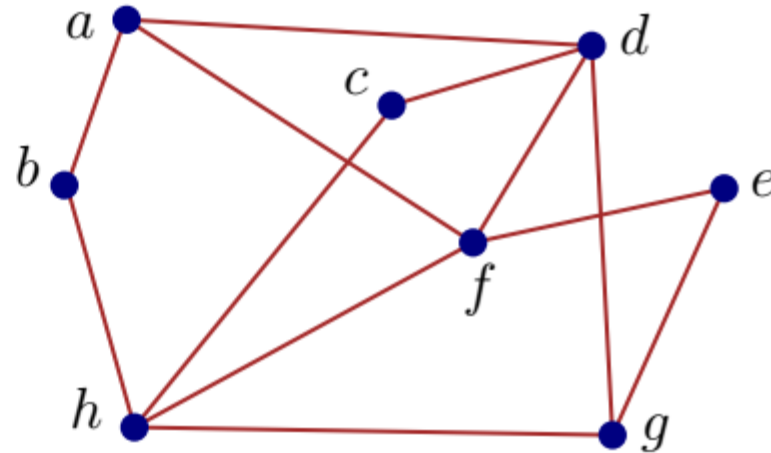
Undirected Simple Graphs

$$G = (V, E)$$

V is set of vertices

E is set of edges

(2-subsets of V , unordered pairs)



Types of Graphs: Undirected Simple Graphs

Let $G = (V, E)$ be a simple graph

Order of the graph is number of vertices in it, $|V| = n$

Size of the graph is number of edges in it, $|E|$

ICP 14-01

What is the maximum possible size of G of order n ?

$$\binom{n}{2}$$

Types of Graphs: PseudoGraphs and Multigraphs

Applications require different types of graphs

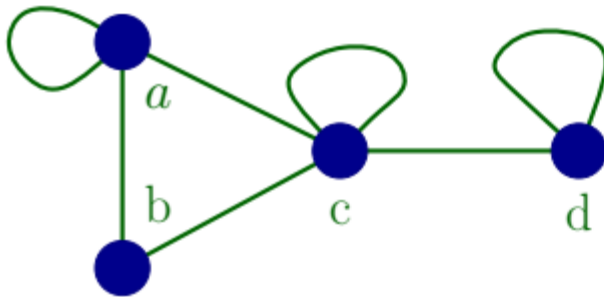
PseudoGraphs

$$G = (V, E)$$

V is set of vertices

E is set of edges

(self loops allowed)



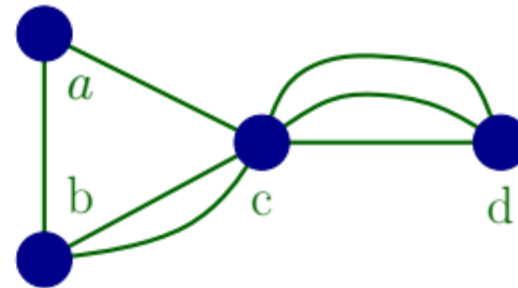
Multigraphs

$$G = (V, E)$$

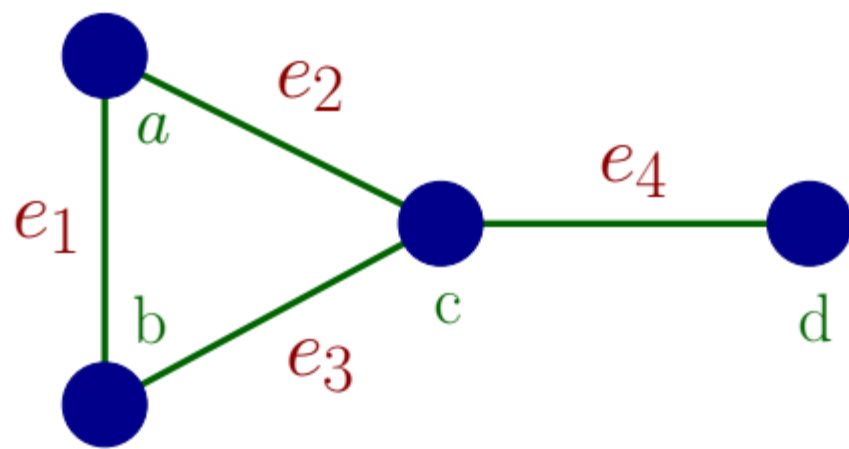
V is set of vertices

E is multi-set of edges

may have self loops too

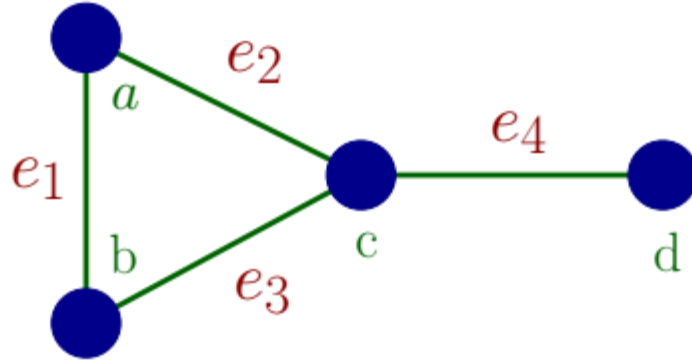


Graph Terminology: Incidence



- $e_1 = (a, b)$: a and b are endpoints of e_1 ▷ slight abuse of notation
- a and b are **adjacent**
- e_1 is incident to a and b

Graph Terminology: Degree



Degree of a vertex is the number of edges incident on it

The number of vertices adjacent to a vertex

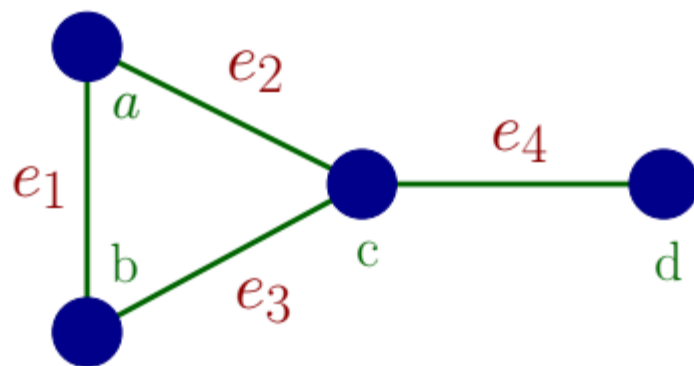
Denoted by $\deg(v)$

ICP 14-02

What is the min and max possible degree of a vertex?

$$0 \leq \deg(v) \leq n - 1$$

Graph Terminology: Degree



In this graph

ICP 14-03

$\deg(a) = ?$

ICP 14-04

$\deg(d) = ?$

ICP 14-05

$\deg(b) = ?$

ICP 14-06

$\deg(c) = ?$

Types of Graphs: Directed Graphs (digraphs)

Applications require different types of graphs

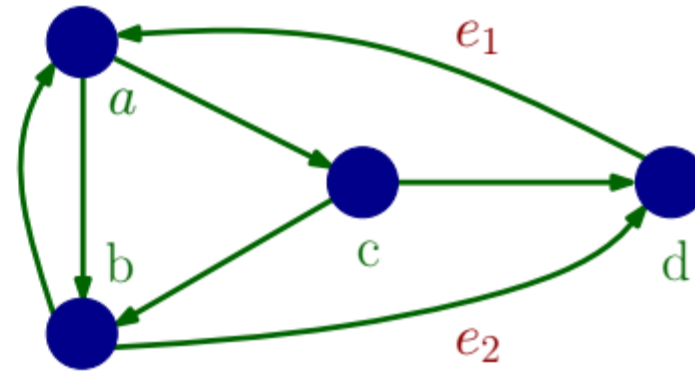
Directed Graphs (digraphs)

$$G = (V, E)$$

V is set of vertices

E is set of edges (ordered pairs)

$$E \subseteq V \times V \text{ (ordered pairs)}$$



Types of Graphs: Directed Graphs (digraphs)

Let $G = (V, E)$ be a simple digraph

Order of the graph is number of vertices in it, $|V| = n$

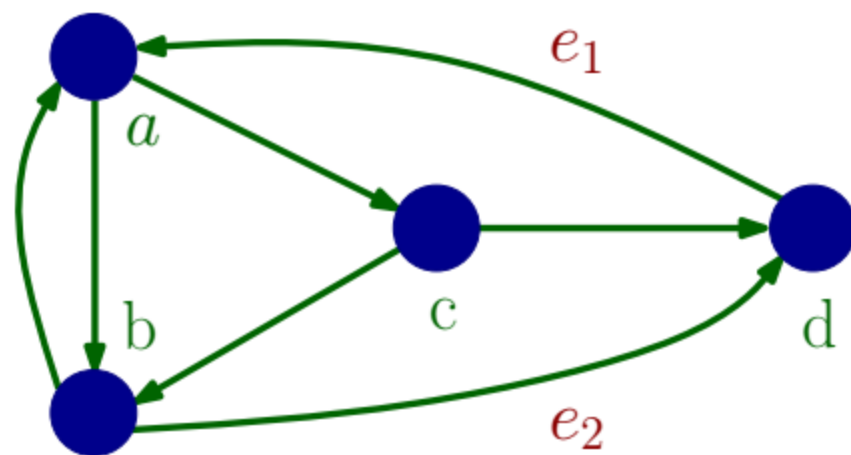
Size of the graph is number of edges in it, $|E|$

ICP 14-07

What is the maximum possible size of G of order n ?

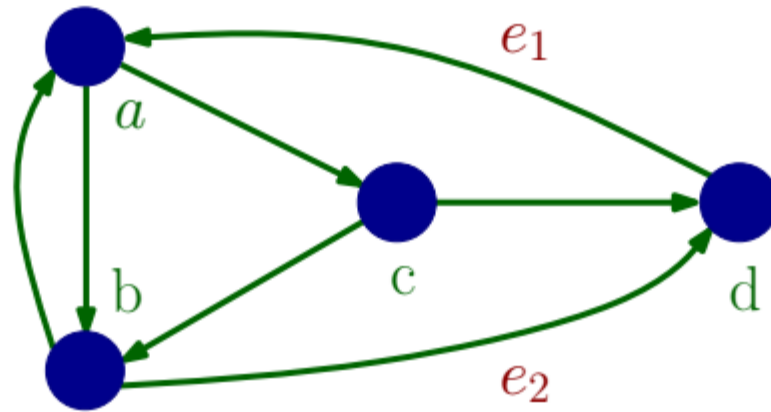
$$n(n - 1)$$

Graph Terminology: Directed Edges



- $e_1 = (d, a) \neq (a, d)$
- d is source of e_1 and a is target of e_1

Graph Terminology: In-degree and Out-degree



in-degree of a vertex is the number of (directed) edges incoming into it

The number of edges with this vertex as a target

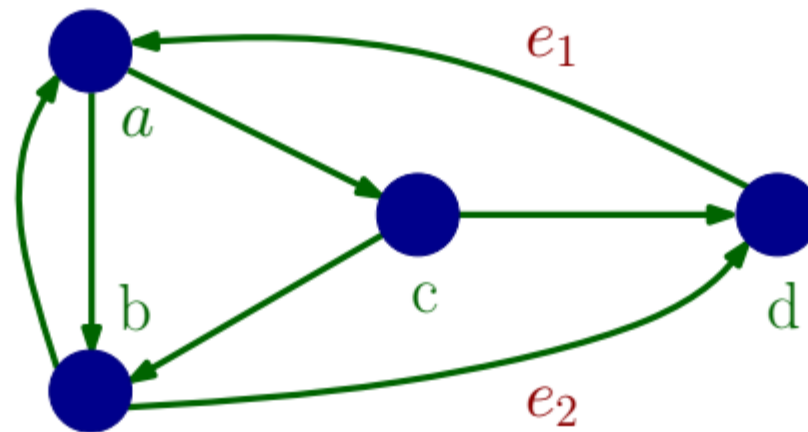
Denoted by $\deg^-(v)$ or $d^-(v)$

ICP 14-08

What is the min and max possible in-degree of a vertex?

$$0 \leq \deg^-(v) \leq n - 1$$

Graph Terminology: In-degree and Out-degree



out-degree of a vertex is the number of (directed) edges outgoing from it

The number of edges with this vertex as a source

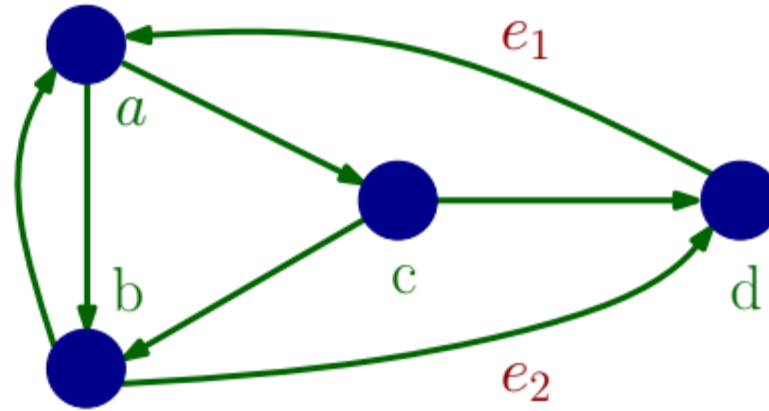
Denoted by $\deg^+(v)$ or $d^+(v)$

ICP 14-09

What is the min and max possible out-degree of a vertex?

$$0 \leq \deg^+(v) \leq n - 1$$

Graph Terminology: In-degree and Out-degree



In this graph

ICP 14-10

$$\deg^+(a) = ?$$

ICP 14-14

$$\deg^-(a) = ?$$

ICP 14-11

$$\deg^+(d) = ?$$

ICP 14-15

$$\deg^-(d) = ?$$

ICP 14-12

$$\deg^+(b) = ?$$

ICP 14-16

$$\deg^-(b) = ?$$

ICP 14-13

$$\deg^+(c) = ?$$

ICP 14-17

$$\deg^-(c) = ?$$

Graph Representation: Adjacency Matrix

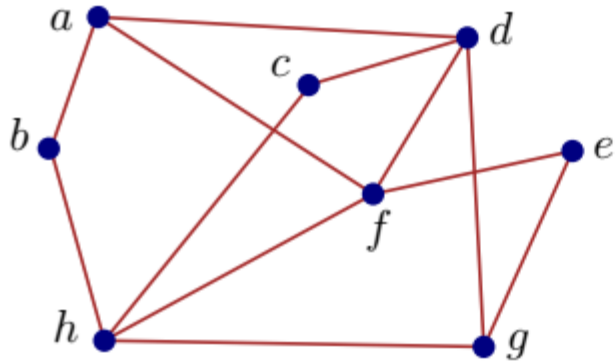
Undirected Simple Graphs

$$G = (V, E)$$

V is set of vertices

E is set of edges

(unordered pairs (2-subsets) of V)



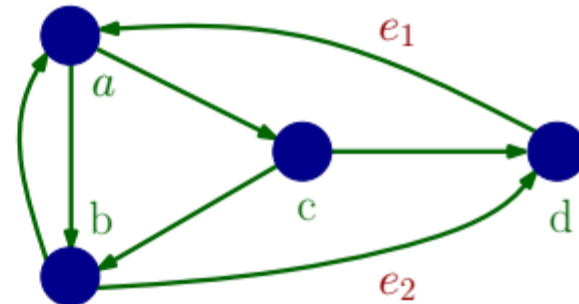
Directed Graphs (digraphs)

$$G = (V, E)$$

V is set of vertices

E is set of edges

(ordered pairs of V)

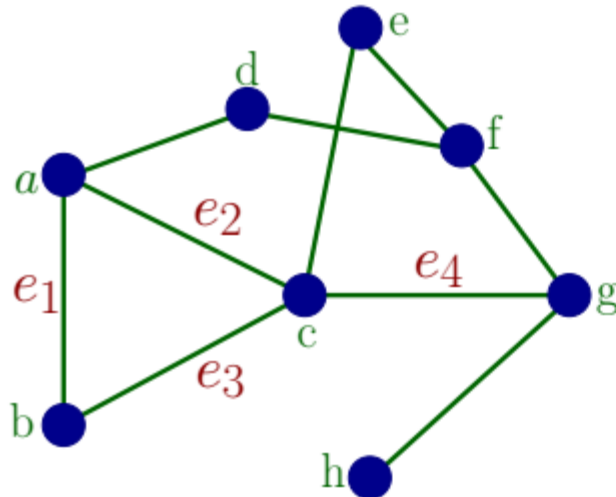


Undirected Graph Representation: Adjacency Matrix

Represent undirected $G = (V, E)$ with an **adjacency matrix** A_G

- Fix an arbitrary ordering of V
- One row for each vertex in V
- One column for each vertex in V

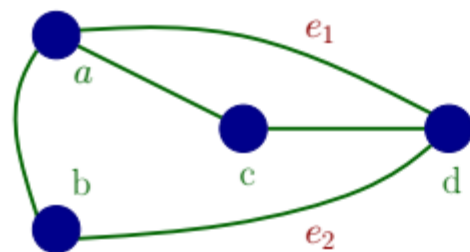
$$A_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{if } (v_i, v_j) \notin E \end{cases}$$



$A_G =$

	a	b	c	d	e	f	g	h
a	0	1	1	1	0	0	0	0
b	1	0	1	0	0	0	0	0
c	1	1	0	0	1	0	1	0
d	1	0	0	0	0	1	0	0
e	0	0	1	0	0	1	0	0
f	0	0	0	1	1	0	1	0
g	0	0	1	0	0	1	0	1
h	0	0	0	0	0	0	1	0

Undirected Graph Representation: Adjacency Matrix



$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

Let A be the adjacency matrix of a simple graph graph on n vertices.

ICP 14-22

How many entries are there in A ?

ICP 14-23

How many 1's are there in A ?

ICP 14-24

How many 0's are there in A ?

ICP 14-25

What are diagonal entries of A ?

ICP 14-26

How many 1's are there in row corresponding to vertex v ?

ICP 14-27

How many 1's are there in column corresponding to v ?

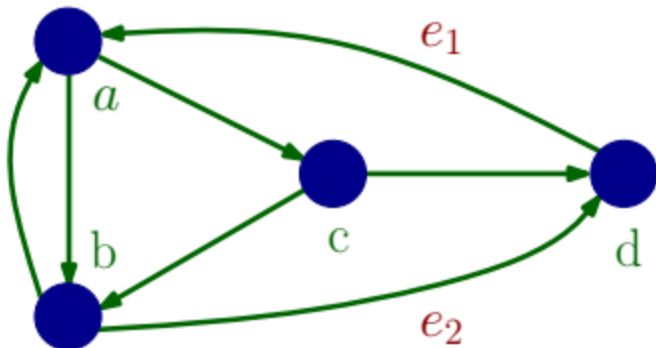
Directed Graph Representation: Adjacency Matrix

Digraph $G = (V, E)$ is a relation on V

Represent G with an **adjacency matrix** A_G

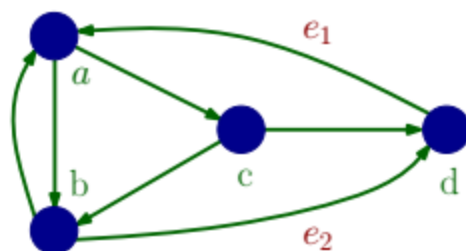
- Fix an arbitrary ordering of V
- One row for each vertex in V
- One column for each vertex in V

$$A_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{if } (v_i, v_j) \notin E \end{cases}$$



$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Directed Graph Representation: Adjacency Matrix



$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Let A be the adjacency matrix of a simple digraph graph on n vertices.

ICP 14-28 How many entries are there in A ?

ICP 14-29 How many 1's are there in A ?

ICP 14-30 How many 0's are there in A ?

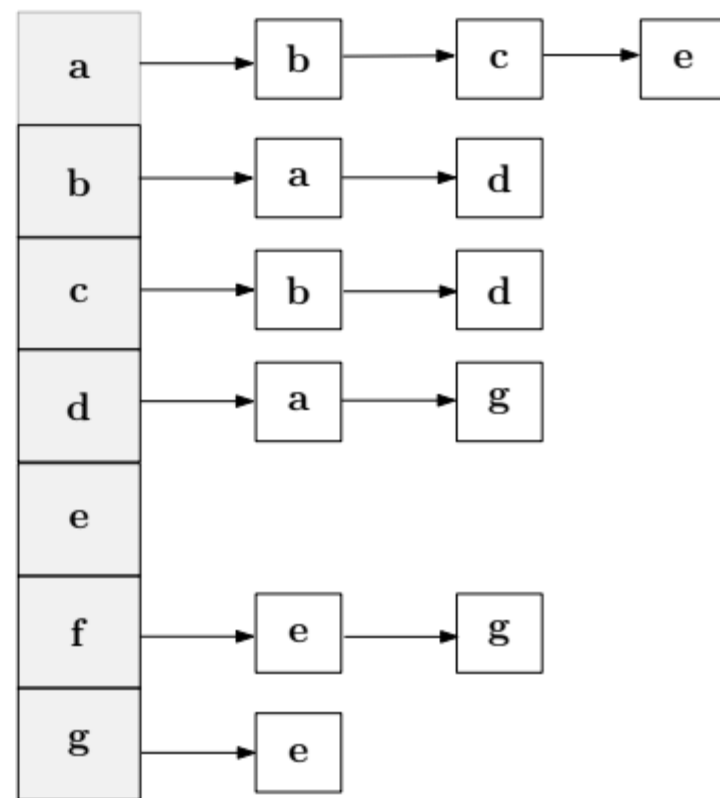
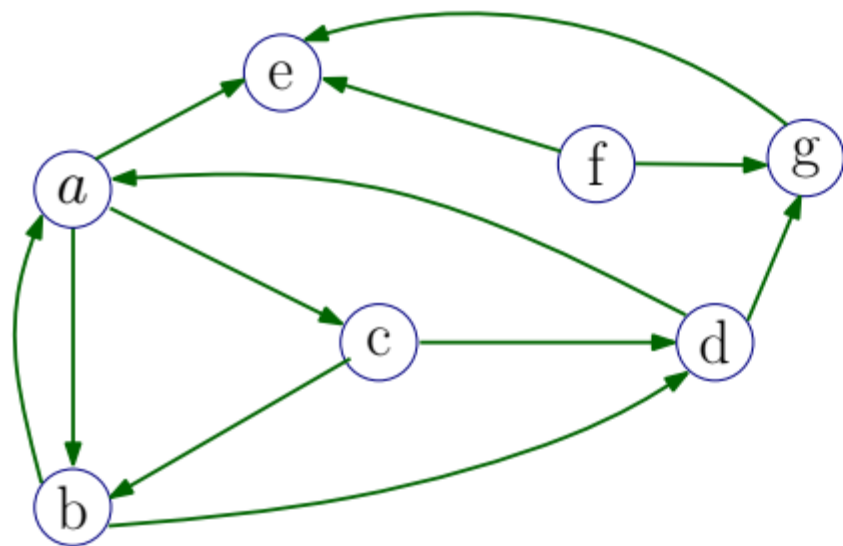
ICP 14-31 What are diagonal entries of A ?

ICP 14-32 How many 1's are there in row corresponding to vertex v ?

ICP 14-33 How many 1's are there in column corresponding to v ?

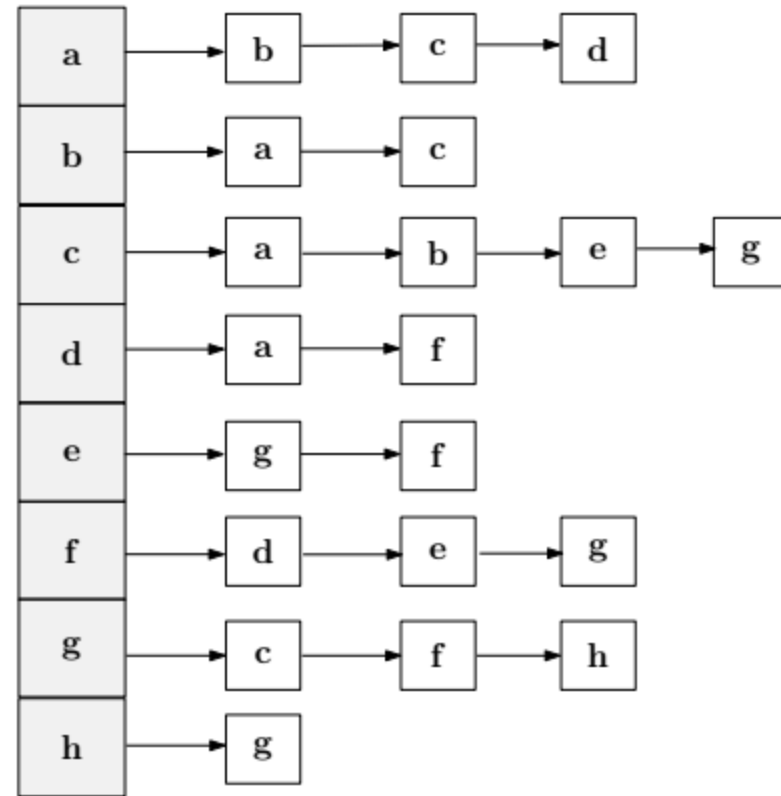
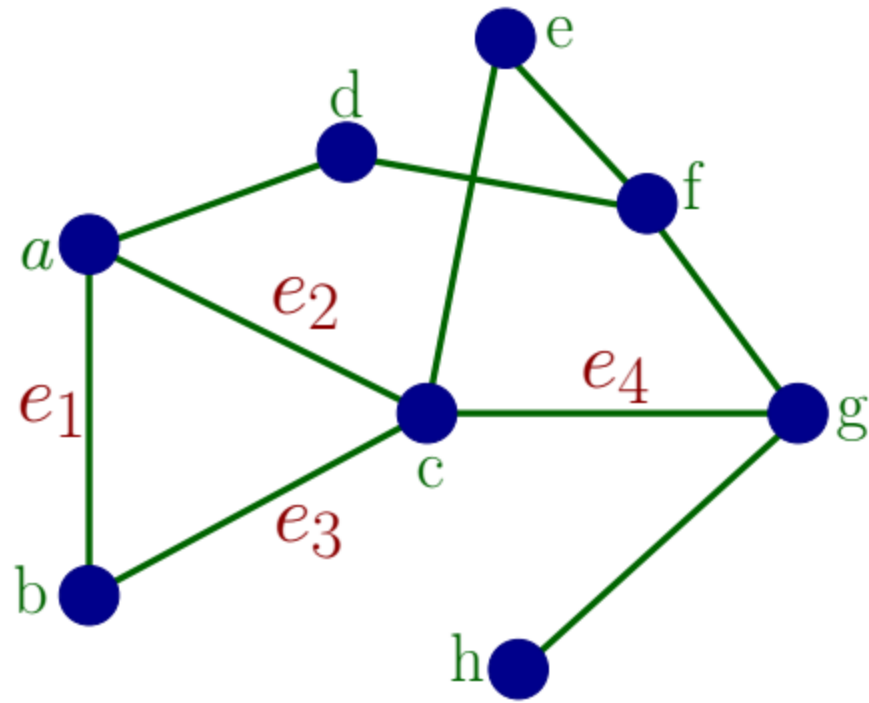
Graph Representation: Adjacency List

Represent digraph by listing neighbors of each vertex



Graph Representation: Adjacency List

Represent undirected graph by listing neighbors of each vertex



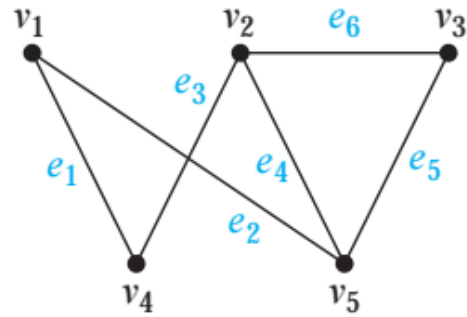
Incidence Matrices

Another common way to represent graphs is to use **incidence matrices**. Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges of G . Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix $\mathbf{M} = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

EXAMPLE 6 Represent the graph shown in Figure 6 with an incidence matrix.

Solution: The incidence matrix is

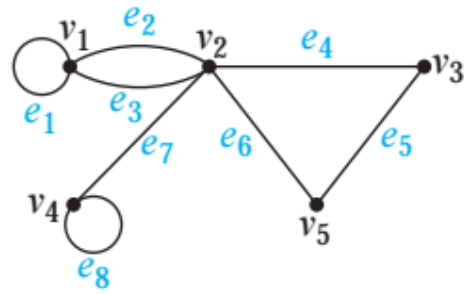


$$\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} e_1 \quad e_2 \quad e_3 \quad e_4 \quad e_5 \quad e_6 \\ \left[\begin{array}{cccccc} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{array} \right] \end{array}.$$

FIGURE 6 An Undirected Graph.

Incidence matrices can also be used to represent multiple edges and loops. Multiple edges are represented in the incidence matrix using columns with identical entries, because these edges are incident with the same pair of vertices. Loops are represented using a column with exactly one entry equal to 1, corresponding to the vertex that is incident with this loop.

EXAMPLE 7 Represent the pseudograph shown in Figure 7 using an incidence matrix.



Solution: The incidence matrix for this graph is

$$\begin{array}{c}
 \begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array}
 \begin{array}{c} e_1 \quad e_2 \quad e_3 \quad e_4 \quad e_5 \quad e_6 \quad e_7 \quad e_8 \\
 \left[\begin{array}{cccccccc}
 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\
 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\
 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0
 \end{array} \right]
 \end{array}
 \end{array}$$

FIGURE 7
A Pseudograph.